

Chapter 1.7 - Group Actions

Emil Marinov

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1 Problem 18

Let H be a group acting on set A . Prove that the relation $\sim$ on A defined by

$$a \sim b \quad \text{iff } a = hb \quad \text{for some } h \in H \quad (1)$$

is an equivalence relation. (For each $x \in A$ the equivalence class of x under $\sim$ is called the **orbit** of x under the action of H . The orbits under the action of H partition the set A .)

To show an equivalence, reflexivity, symmetry, and transitivity must be shown.

Reflexivity states that $(a, a) \in \rho \forall a$, where ρ is the relation.

$$a = ea \quad (2)$$

As $e \in H$, there exists an $h = e \in H$ such that $a = ha$. Thus, $a \sim a$.

Symmetry. The relation on a and b is symmetric iff $(a, b) \in \rho \implies (b, a) \in \rho$. Let $a = hb$.

$$\begin{aligned} a &= hb \\ h^{-1}a &= (h^{-1}h)b \\ h^{-1}a &= b \end{aligned} \quad (3)$$

$h^{-1} \in H$ and there exists a $g = h^{-1} \in H$ such that $b = ga$. Thus symmetry has been shown.

Transitivity. The relation $\sim$ is transitivity iff $(x, y) \in \rho, (y, z) \in \rho \implies (x, z) \in \rho$. Let $a = h_1b$ and $b = h_2c$, for $h_1, h_2 \in H$.

$$\begin{aligned} a &= h_1b \\ a &= h_1(h_2c) \\ a &= (h_1h_2)c \\ a &= h_3c \quad \text{Let } h_3 = h_1h_2 \in H \\ a &\sim c \end{aligned} \quad (4)$$

The group operation h_1h_2 yields an element $h_3 \in H$ due to closure.

The relation $\sim$ on set A defined by $a \sim b$ iff $a = hb$ for some $h \in H$ has been shown to be reflexive, symmetric, and transitive. Thus, it has been proven to be an equivalence relation. $\square$

The same goes for $y \in A$, or any other element in A . The orbits of different elements $x \in A$ under the action of H are disjoint, as indicated by the equivalence relation. If they were not disjoint, then the equivalence relation indicates they share ALL elements together i.e. being the same orbit.

2 Problem 20

Let G be a group and let G act upon itself by left conjugation, so each $g \in G$ maps G to G by

$$x \rightarrow gxg^{-1} \quad (5)$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself (i.e., is an **automorphism** of G - cf. Exercise 20, Section 6). Deduce that x and gxg^{-1} have the same order for all x in G and that for any subset A of G , $|A| = |gAg^{-1}|$ (here $gAg^{-1}|a \in A$).

To show that the left conjugation on G is an isomorphism, it must be shown that 1) it is a homomorphism, 2) it is injective, and 3) it is surjective.

Show Homomorphic

$$\begin{aligned} \varphi(xy) &= g(xy)g^{-1} \\ &= gxyg^{-1} \quad \text{By associativity.} \\ &= gxg^{-1}gyg^{-1} \\ &= (gxg^{-1})(gyg^{-1}) \\ &= \varphi(x)\varphi(y) \end{aligned} \quad (6)$$

Show Injective

Here, it is needed to be shown that $\varphi(x) = \varphi(y) \implies x = y$.

$$\begin{aligned} \varphi(x) &= gxg^{-1} = gyg^{-1} = \varphi(y) \\ gxg^{-1}g &= gyg^{-1}g \\ gx &= gy \\ g^{-1}gx &= g^{-1}gy \\ x &= y \end{aligned} \quad (7)$$

Show Surjective

To show surjectivity, it must be shown that each element from the codomain has an element in the domain that maps to it: $\forall y \in G, \exists x \in G$ s.t. $\varphi(x) = y$.

$$\begin{aligned} y &= \varphi(x) = gxg^{-1} \\ \implies x &= g^{-1}yg \end{aligned} \tag{8}$$

This holds for all $g \in G$. Thus, it is surjective.

Order

Let n be the order of x , i.e. $x^n = 1$. Let m be the order of $\varphi(x)$, i.e. $\varphi(x)^m = 1$.

$$\begin{aligned} \varphi(x)^m &= (gxg^{-1})^m = gx^mg^{-1} = e \\ x^m &= g^{-1}eg = g^{-1}g = e \end{aligned} \tag{9}$$

If $x^n = e$, then $gx^n g^{-1} = e$. This implies $|\varphi(x)| = m \leq n$. If $gx^m g^{-1} = e$, then $x^m = e$. This implies $|x| = n \leq m$.

Because $m \leq n$ and $n \leq m$, $m = n$. Thus, the order of the automorphism is the same as that for the input group element.

Order 2

gAg^{-1} is bijective with A . Thus, $|A| = |gAg^{-1}|$.